Library Management System Using JDBC Statement

Project Structure

```
LibraryManagementStatement
■
■■■ src
■    ■■■ Book.java
■    ■■■ BookDAO.java
■    ■■■ Main.java
■
■■■ mysql-connector-java.jar
```

Database

```
create database librarydb;

use librarydb;

create table books
(
    book_id int primary key,
    title varchar(50),
    author varchar(50),
    category varchar(30),
    price double
);
```

Book.java

```
public class Book {

    private int bookId;
    private String title;
    private String author;
    private String category;
    private double price;

    public Book(int bookId,String title,String author,String category,double price)
    {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.category = category;
        this.price = price;
    }

    public int getBookId(){ return bookId; }
    public String getTitle(){ return title; }
    public String getAuthor(){ return author; }
    public String getCategory(){ return category; }
    public double getPrice(){ return price; }
}
```

BookDAO.java

```
import java.sql.*;

public class BookDAO {

 Connection con;

 public BookDAO() throws Exception {

  Class.forName("com.mysql.cj.jdbc.Driver");

  con = DriverManager.getConnection(
   "jdbc:mysql://localhost:3306/librarydb",
   "root",
```

```
  "root");
 }

 public void insertBook(Book b) throws Exception {

  Statement st = con.createStatement();

  String sql =
  "insert into books values("
  + b.getBookId() + ",'"
  + b.getTitle() + "','"
  + b.getAuthor() + "','"
  + b.getCategory() + "',"
  + b.getPrice() + ")";

  st.executeUpdate(sql);
 }

 public void updatePrice(int bookId,double price)
 throws Exception {

  Statement st = con.createStatement();

  String sql =
  "update books set price=" + price +
  " where book_id=" + bookId;

  st.executeUpdate(sql);
 }

 public void deleteBook(int bookId)
 throws Exception {

  Statement st = con.createStatement();

  String sql =
  "delete from books where book_id="
  + bookId;

  st.executeUpdate(sql);
 }

 public void displayCategory(String category)
 throws Exception {

  Statement st = con.createStatement();

  String sql =
  "select * from books where category='"
  + category + "'";

  ResultSet rs = st.executeQuery(sql);

  while(rs.next()) {

   System.out.println(
   rs.getInt(1)+" "+
   rs.getString(2)+" "+
   rs.getString(3)+" "+
   rs.getString(4)+" "+
   rs.getDouble(5));
  }
 }
}
```

Main.java

```
import java.util.Scanner;

public class Main {

 public static void main(String[] args)
 throws Exception {

  Scanner sc = new Scanner(System.in);
  BookDAO dao = new BookDAO();
```

```
  while(true) {

   System.out.println("1 Insert Book");
   System.out.println("2 Update Price");
   System.out.println("3 Delete Book");
   System.out.println("4 Display By Category");
   System.out.println("5 Exit");

   int ch = sc.nextInt();

   switch(ch) {

    case 1:
      // Insert Logic
      break;

    case 2:
      // Update Logic
      break;

    case 3:
      // Delete Logic
      break;

    case 4:
      // Display Logic
      break;

    case 5:
      System.exit(0);
   }
  }
 }
}
```

Viva Questions

```
Statement:
Statement st = con.createStatement();

PreparedStatement:
PreparedStatement ps = con.prepareStatement(sql);

CallableStatement:
CallableStatement cs =
con.prepareCall("{call ProcedureName(?)}");

Statement -> SQL as String
PreparedStatement -> SQL with ? placeholders
CallableStatement -> Stored Procedures
```